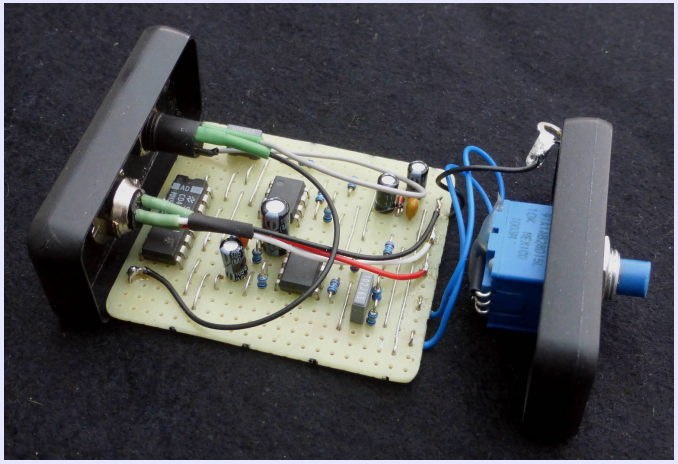
ZOOM F3 - Ultrasonic Heterodyne Monitor

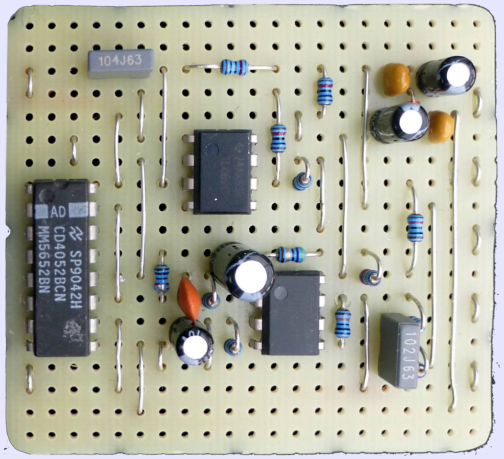
• Enclosure is Hammond: 1455C802BK •

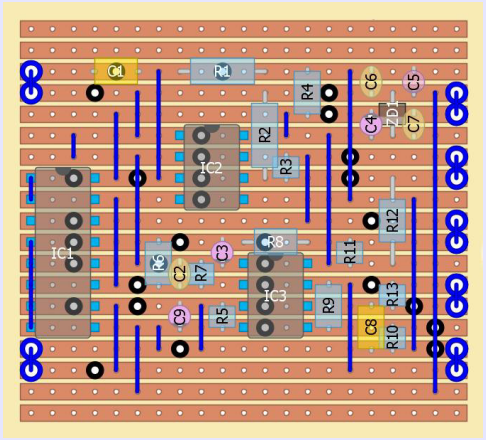
- **Frequency shifts ultrasound to audible audio range**
- **Ultrasonic input from channel 1 line output**
- **Phantom powered from channel 2 XLR input**
- **Monitor audio out from channel 2 headphone output**

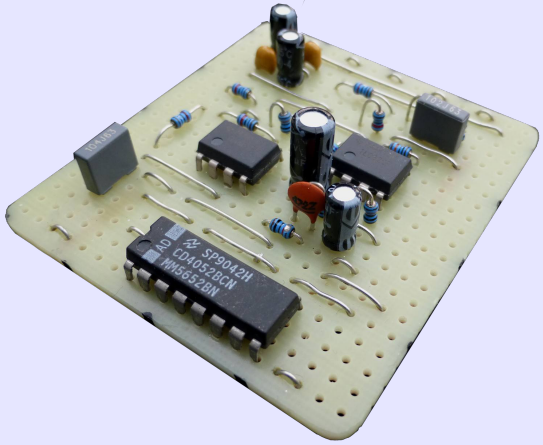






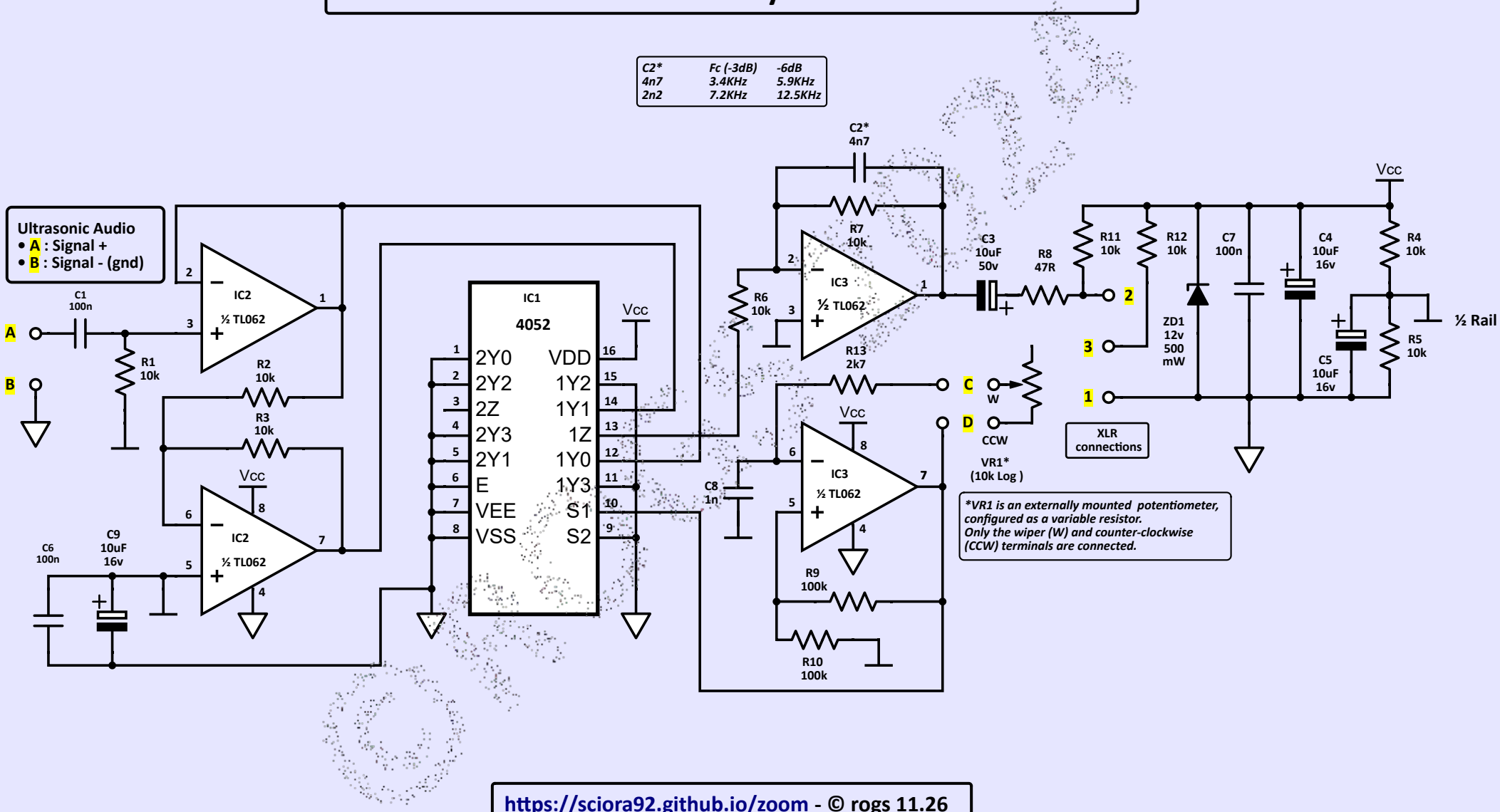






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• Zoom F3 Ultrasonic Heterodyne Monitor Schematic •



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Zoom F3 Heterodyne Monitor - Circuit description

Overview:

The Zoom F3 field recorder is not primarily designed to record ultrasound. However, with a 192KHz maximum sampling rate - and a useful c.70KHz upper frequency response throughout the signal path - it is capable of recording significant proportions of the ultrasound created by the echolocation calls of many species of bat.
The 32 bit float format is also a very useful feature when attempting to record these signals, which have such an unpredictable dynamic range.

There does remain the issue of real time monitoring of these ultrasonic signals. The vast majority are beyond the range of human hearing, and so it is useful to be able to use a simple heterodyne monitor to bring them within a more useful audio range, in real time.

This project allows for the construction of such a device, using a simple balanced switched mixer created from a single low cost CMOS analogue multiplexer and two TL062 op-amps.

The project is powered from the phantom power available at the F3 XLR inputs. An ultrasonic microphone is connected to one of the F3 XLR inputs, to actually record the ultrasound. The 'line out' socket from this channel is connected to the input of this heterodyne monitor. The monitor circuitry is powered from the phantom power of the second F3 XLR input. The heterodyne monitor audio output is connected to pin 2 of the same XLR socket, and the output is taken from the F3 headphone socket of this second channel, to enable real time audio monitoring of the ultrasound being detected and recorded on the primary channel.

Circuit description:

The ultrasonic signal from the line output of the primary recording channel is connected via C1 to the pin3 non-inverting input of IC2, a TL062 op-amp configured as a unity gain non-inverting buffer. The op-amp output (pin1) is connected directly to the inverting input to achieve this.
The output of this op-amp is connected, via R2, to the pin 6 inverting input of the second half of this same TL062 device. The feedback resistor R3 allows for a unity gain output signal of inverted polarity to appear at the output (pin 6) of this second op-amp.

The opposite polarity outputs of both amplifiers are presented to pins 12 and 14, the two inputs of one channel of IC1, a 4052 CMOS dual analogue multiplexer.
(The second channel of this device is not used, and all the unused inputs are connected to 0v (gnd))
The 4052 acts as a double pole analogue switch. The output of this switch at pin 13 routes the two opposing op-amp output signals to the pin 2 inverting input of one half of a second TL062 op-amp, IC3, via R6.

The rate at which these two signals are switched is determined by the rate of the local oscillator, based around pins 5,6 & 7 of IC3.
This second half of IC3 is configured as an adjustable relaxation oscillator. R9 & R10 act as the Schmitt trigger, with R13 and C8 providing the timing components. The external variable resistor VR1 allows for the frequency to be adjusted between c.25KHz and c.65KHz. This range should allow selection of the local oscillator frequency to be close to the incoming ultrasound frequency, and so allow for the 'heterodyne' output to act as an audible monitor signal.

The multiplexer switching action performs a multiplication function of the op-amp and local oscillator signals, the output of which is a signal at the difference frequency of these inputs.
(Some notes on the details of the maths behind this mixing function can be found here: HERE)

The resulting frequency shifted audio signal is passed through the first half of IC3, which is configured as a unity gain inverting amplifier. The bandwidth of this amplifier is restricted by C2, which acts as a low pass filter to attenuate residual ultrasound from the mixing process. The desired upper frequency limits of the output signal can be varied, simply by changing the value of C2. Typical values are shown on the schematic.

The final audio output signal is fed from IC3 pin 1 - via C3 and R8 - to pin 2 of the second channel F3 XLR input.

Power to the device is derived from the 48v phantom power supply of the selected F3 channel. This power is taken from both sides of the power supply - via R11 and R12 - to be regulated by the 12v Zener diode ZD1.
C7 and C4 serve to decouple the 12v DC output (Vcc) which is used to power all the ICs.
R4 and R5 serve to act as a voltage divider - decoupled by C5, C6 & C9 - to provide a 'half rail' supply to the op-amp inputs. This supply serves to bias the op-amps, and enable correct operation from a single 12v supply.

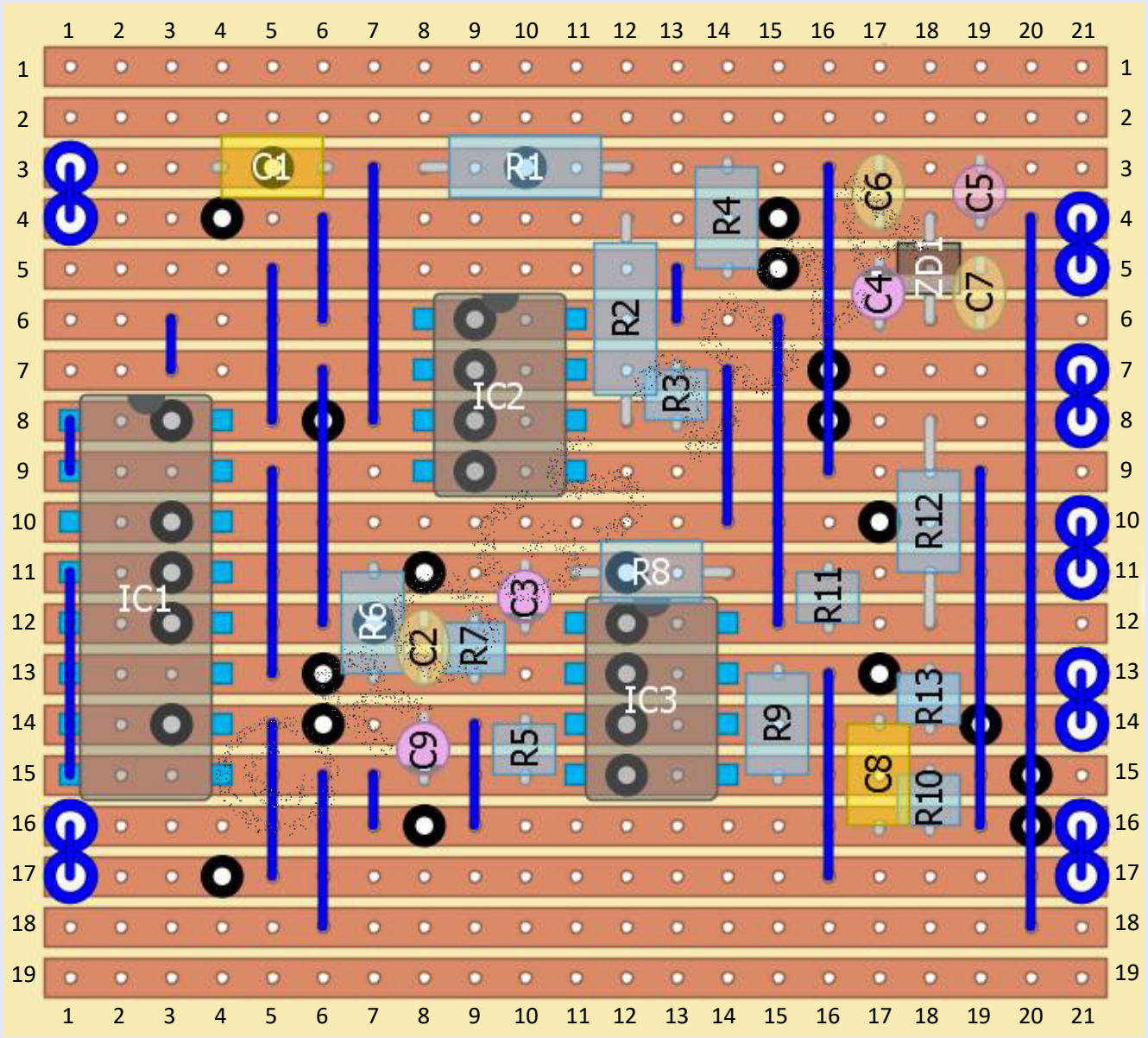
• ZOOM F3 HETERODYNE MONITOR - STRIPBOARD LAYOUT •

• *A & B are to tip and sleeve of 3.5mm TRS jack socket*

• *'Double pad' connection points allow for more robust cable terminations.*

CONNECTIONS :

- **A :** Ultrasonic Audio In +
- **B :** Ultrasonic Audio In – (gnd)
- **C :** Ext. Control Pot. (W)
- **D :** Ext. Control Pot. (CCW)
- **1 :** Mini XLR pin 1 (gnd)
- **2 :** Mini XLR pin 2
- **3 :** Mini XLR pin 3



• *Pins 1,2,4,5,6,7 & 8 of IC1 are linked with solder bridges on the track side of the stripboard.*

This allows the board to be kept to a size suitable for mounting into a Hammond 1455C802 enclosure.

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PARTS LIST :

Semiconductors:
- IC1 : CD4052 or HEF4052
- IC2 & IC3 : TL062
- ZD1 12v 500mW Zener diode

Resistors: *(MF12 series)*

- R1 10k
- R2 10k
- R3 10k
- R4 10k
- R5 10k
- R6 10k
- R7 10k
- R8 47R
- R9 100k
- R10 100k
- R11 10k
- R12-10k
- R13 2k7

Capacitors:

- C1 100n
- C2 4n7
- C3 10uF50v
- C4 10uF16v
- C5 10uF16v
- C6 100n
- C7 100n
- C8 1n
- C9 10uF 16v

- *C1 & C8 are PET 63v 0.2"*
- *C2, C6 & C7 are MLCC 0.1"*
- *C3,C4,C5 & C9 are electrolytics*

Stripboard :

- FC2 or FC4 (*KEMO E005*)
- 19 (copper tracked) rows
- 21 columns

- Wire links (TCW 25) total : 26 (Blue lines on drawing)
- 'Spin off' track cuts total : 33 (Black and white dots on drawing)